University of Zagreb
Faculty of Mechanical Engineering and Naval Architecture

BACHELOR'S THESIS

Andrija Adamović

Zagreb, 2026

University of Zagreb

Faculty of Mechanical Engineering and Naval Architecture

BACHELOR'S THESIS

Mentor: Student:

Prof. dr. sc. Andrej Jokić Andrija Adamović

Zagreb, 2026

I hereby declare that I have written this thesis independently, using the knowledge acquired during my studies and the literature cited in the references.

TODO: Zahvale

Andrija Adamović

SVEUČILIŠTE U ZAGREBU
FAKULTET STROJARSTVA I BRODOGRADNJE

Središnje povjerenstvo za završne i diplomske ispite
Povjerenstvo za završne ispite studija mehatronika i robotika

Sveučilište u Zagrebu Fakultet strojarstva i brodogradnje	
Datum	Prilog
Klasa: 602-01/26-01/3	
Ur.broj: 15-31000-26-	

ZAVRŠNI ZADATAK

Student: **Andrija Adamović** JMBAG: 0035248277

Naslov rada na hrvatskom jeziku: **Willemsova fundamentalna lema s primjenama**

Naslov rada na engleskom jeziku: **Willems' Fundamental Lemma with Applications**

Opis zadatka:

Classical approach to simulation and controller design for dynamical systems is a model-based approach. A model of system's dynamics is obtained either directly by applying laws of physics on the considered system or indirectly via system identification. The latter is a data-based approach and the data, which consists of recorded input-output trajectories of the system, is used in a suitably defined optimization problem to obtain a model.
Alternative to model-based approaches is the direct data-based approach where simulations and/or controller synthesis is performed directly from data, completely avoiding a model. Such approach has gained significant attention in the control systems community over the past years. There are several reasons for that and the obvious one is that modern IT technologies enable collecting (sensing/recording) of large data sets in an inexpensive way. From a scientific point of view, the direct data-based approach is attractive as it can offer exact results with robustness guarantees, similar to the classical model-based approach.
One of the key results that has inspired and enabled a set of methods in direct data-based approach is the so-called Willems' fundamental lemma. Loosely speaking, this lemma states that all possible trajectories of a linear time invariant (LTI) dynamical system belong to an image space of a suitable defined data-based matrix.
The main goal of this thesis is to summarize dominant state-of-the-art approaches to data-based simulation and controller design and to suitably illustrate them on several examples.
The thesis should contain the following:

- Presentation of the Willems' fundamental lemma together with discussion of its origins and meaning.
- Literature overview of the state-of-the-art results on data-based approaches which are based on the Fundamental lemma.
- On a set of suitably selected and designed examples illustrate some of the data-based approaches for simulation and control of LTI dynamical systems.
- Using suitable simulation tools (e.g., MATLAB/Simulink) to investigate applicability of the Fundamental lemma on nonlinear systems.

The thesis must contain the list of used literature as well as any assistance received.

Zadatak zadan:	Datum predaje rada:	Predviđeni datumi obrane:
22. 4. 2026.	**2. rok:** 15. i 16. 7. 2026. **3. rok:** 16. i 17. 9. 2026.	**2. rok:** 21. - 23. 7. 2026. **3. rok:** 22. - 24. 9. 2026.
Zadatak zadao:		Predsjednik Povjerenstva:
Prof. dr. sc. Andrej Jokić		izv. prof. dr. sc. Petar Ćurković

Contents

List of Figures

List of Tables

LIST OF SYMBOLS

Symbol	Description	Unit
x	Example variable	/

LIST OF ABBREVIATIONS

Abbreviation	Description
LTI	Linear time-invariant

SAŽETAK

Ovdje upišite sažetak rada na hrvatskom jeziku.

KLJUČNE RIJEČI: ključna riječ jedan; ključna riječ dva

ABSTRACT

Write the thesis abstract in English here.

Keywords: keyword one; keyword two; keyword three

1 Introduction

Classical control systems almost always use a system model to synthesise a control algorithm.

1.1 Motivation

Describe the background and motivation for the thesis.

1.2 Literature Overview

2 The Fundamental Lemma

2.1 Theoretical Background

The goal of this section is to provide introduction to mathematical concepts needed for understanding the rest of the thesis.

2.1.1 LTI Dynamical Systems

Although Willems' Fundamental Lemma stems from the behavioural approach to sytems theory, in this thesis it is formulated using the classical state-space LTI system formulation. This formulation is simpler, yet sufficient for the further chapters of the thesis.

Definition 1. *A discrete-time linear time-invariant (LTI) system is a dynamical system whose dynamics are described by:*

$$x(t+1) = Ax(t) + Bu(t) \tag{2.1a}$$

$$y(t) = Cx(t) + Du(t) \tag{2.1b}$$

Where $t \in \mathbb{Z}_{\geq 0}$ is the time axis, $x(t) \in \mathbb{R}^n$ is the state, $u(t) \in \mathbb{R}^m$ is the input and $y(t) \in \mathbb{R}^p$ is the output of the system.
The true state space dimension is n and system matrices $A \in M_n(\mathbb{R})$, $B \in M_{n\times m}(\mathbb{R})$, $C \in M_{p\times n}(\mathbb{R})$, $D \in M_{p\times m}(\mathbb{R})$.

In our data-based approach, those matrices (i.e. the system model) are generally unknown. However, an upper bound on state space dimension is given $N \geq n$. The concept of a system trajectory is defined now.

Definition 2. *A sequence of ordered triples $(u(t), x(t), y(t))_{t=0}^{\infty}$ is called an input-state-output trajectory if*

$$\begin{bmatrix} x(t+1) \\ y(t) \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} x(t) \\ u(t) \end{bmatrix}$$

We can be deduce that, from the linearity of (2.1), any linear combination of input-state-ouput trajectories is itself an input-state-output trajectory i.e. the set of all such trajectories spans a vector space over $\mathbb{R}$. That space is called the *behaviour* of the system. Additionally, from time invariance (2.1) it can be deduced that:

$$(u(t+\tau), x(t+\tau), y(t+\tau))_{t=0}^{\infty}$$

is an input-state-output trajectory for all $\tau \in \mathbb{N}$

Definition 3. *A sequence of ordered pairs $(u(t), y(t))_{t=0}^{\infty}$ is called an input-output trajectory if there exists a sequence of states $x : \mathbb{Z}_{\geq 0} \to \mathbb{R}^n$ such that $(u(t), x(t), y(t))_{t=0}^{\infty}$ is an input-state-output trajectory.*

In practice we are limited by finite computer memory so it is not possible work with infinite sequences of data like the trajectories defined above, so we need to take a snippet of the trajectory relevant to us. Motivated by that we state the next definition.

Definition 4. *Let $i, j \in \mathbb{Z}_{\geq 0}$ such that $i \leq j$. Given an input-state-output trajectory $(u(t), x(t), y(t))_{t=0}^{\infty}$, the sequence $(u(t), x(t), y(t))_{t=i}^{j}$ is called a restricted input-state-output trajectory. Restricted input-output trajectories are defined in the same way.*

2.1.2 The Concepts of Contollability and Observability

In this section the concepts of controllability and observability will be defined. Since, we defined the LTI system (2.1) classically the same will follow for the next definitions. However, although the classical and behavioral notions of controllability and observability are closely related, they are not identified unconditionally. Their equivalence depends on how the behavior is represented. In particular, when considering input-output behavior, equivalence with the classical state-space definitions typically requires a minimal realization; for the full input-state-output behavior, the correspondence is more direct. We will not bother with those comparisons between approaches, more on that topic can be found at (TODO: link za clanak/knjigu).

Definition 5. *A discrete LTI system (2.1) is called controllable if for any initial state $x(0)$ and any desired state x_f, there exists a finite-length sequence of inputs*

$$u(0), u(1), ..., u(N-1)$$

such that the system reaches $x(N) = x_f$. Otherwise, the system is said to be uncontrollable.

In lays terms, it is telling that any state of the system can be reached in finite time by applying finitely many inputs.
We will now state the famous result for testing system controllability stated and proven by Rudolf. E. Kalman. The pair (A, B) is the pair of matrices from (2.1) and n is the true state space dimension of the system.

Theorem 1. *(Kalman controllability theorem) The pair (A, B) is controllable if and only if the controllability block matrix*

$$\mathcal{C} = \begin{bmatrix} B & AB & A^2B & \cdots & A^{n-1}B \end{bmatrix}$$

is of full rank.

Now the concept of observability is defined.

Definition 6. *A discrete LTI system (2.1) is called observable at $t = 0$ if the initial state x(0) can be uniquely reconstructed from a known restricted input-output trajectory. Otherwise, the system is said to be unobservable.*

Using time-invariance of (2.1) it can be shown that if the system is observable at $t = 0$ then for every $t \in \mathbb{Z}_{\geq 0}$ the state $x(t)$ is reconstructible. This result is important to control theory when it comes to state estimation (for example Kalman filter) because it gives a necessary condition for exact state estimation.
Now the test for determining observability will be stated.

Theorem 2. *(Kalman observability theorem) The pair (C, A) is observable if and only if the observability block matrix*

$$\mathcal{O} = \begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^{n-1} \end{bmatrix}$$

is of full rank.

The Kalman criteria for controllability/observability were stated but, there are more different equivalent statements of which the most used ones are the Popov-Belevitch-Hautus (PBH) tests.
From Theorems 1 and 2, a duality between observability and controllability matrices can be seen and it is not hard to show that using matrix operations we have:

$$(A, B) \text{ is controllable} \iff (B^T, A^T) \text{ is observable.}$$

Similarly,

$$(C, A) \text{ is observable} \iff (A^T, C^T) \text{ is controllable.}$$

2.1.3 Persistency of Excitation

For the Willems' Fundamental Lemma to hold we need a so-called persistently exciting input. By that it means a rich and diverse enough input such that all modes of the system are expressed in the output.

Hankel Matrices

For a sequence of vectors from time $t = i$ to $t = j$ we use the following vector notation:

$$s_{[i,j]} = \begin{bmatrix} s(i) \\ s(i+1) \\ \vdots \\ s(j) \end{bmatrix}$$

Where instead of vector s we will use inputs(u), states(x) and outputs(y).
Hankel matrix is a matrix whose entries are constant on the anti-diagonals, contrast to Toeplitz matrix whose entries are constant on the diagonals. Informally, Hankel matrix is a mirror image of a Toeplitz matrix.

Definition 7. *Given an $T \in \mathbb{N}$, lets consider a restricted input-output trajectory $(u_{[0,T-1]}, y_{[0,T-1]})$ of (2.1). Hankel matrices of depth $L \in \{1, ..., T\}$ of these inputs and outputs are given by:*

$$H_L(u_{[0,T-1]}) = \begin{bmatrix} u(0) & u(1) & \cdots & u(T-L) \\ u(1) & u(2) & \cdots & u(T-L+1) \\ \vdots & \vdots & & \vdots \\ u(L-1) & u(L) & \cdots & u(T-1) \end{bmatrix}$$

$$H_L(y_{[0,T-1]}) = \begin{bmatrix} y(0) & y(1) & \cdots & y(T-L) \\ y(1) & y(2) & \cdots & y(T-L+1) \\ \vdots & \vdots & & \vdots \\ y(L-1) & y(L) & \cdots & y(T-1) \end{bmatrix}$$

Generally, we stack input and output Hankel matrices to obtain a single block Hankel matrix used for computation:

$$\begin{bmatrix} H_L(u_{[0,T-1]}) \\ H_L(y_{[0,T-1]}) \end{bmatrix} \in M_{(m+p)L\times(T-L+1)}(\mathbb{R})$$

It is now obvious that every column of the matrices from (7) is a restricted input or output trajectory, depending on which matrix is chosen.
Now a precise definition of persistently exciting input is given.

Definition 8. *Given an sequence of inputs $u_{[0,T-1]}$ and $k \in \{1, ..., T\}$. We say that the given sequence of inputs is persistently exciting of order k if the Hankel matrix $H_k(u_{[0,T-1]})$ is of full row rank.*

We have layed out all the necessary definitions and theorems needed for the statement of the main topic of this thesis, the Lemma.

2.2 The Lemma

The main result of the Lemma is that every restricted trajectory of length L of the system can be obtained by linear combination of finitely many trajectories which span the whole trajectory space i.e. the behaviour.

Theorem 3. *(Willems' Fundamental Lemma) Given a LTI system (2.1), assume the pair (A, B) is controllable and let $(u_{[0,T-1]}, y_{[0,T-1]})$, $T \in \mathbb{N}$ be it's restricted input-output trajectory. Let $L \in \{1, ..., T\}$, if the input $u_{[0,T-1]}$ is persistently exciting of order $N + L$ then: $(\bar{u}_{[0,L-1]}, \bar{y}_{[0,L-1]})$ is a restricted input-output trajectory of (2.1) if and only if*

$$\begin{bmatrix} \bar{u}_{[0,L-1]} \\ \bar{y}_{[0,L-1]} \end{bmatrix} = \begin{bmatrix} H_L(u_{[0,T-1]}) \\ H_L(y_{[0,T-1]}) \end{bmatrix} g$$

for some $g \in \mathbb{R}^{T-L+1}$.

Proof. Solving the difference equations of (2.1) we obtain the following result:

$$y(t) = CA^t x(0) + \left(\sum_{k=0}^{t-1} CA^{t-k-1} Bu(k) \right) + Du(t)$$

where $x(0) \in \mathbb{R}^n$ is the initial state. Generally, this could be adapted to any initial state $x_{ini} = x(t_{ini})$ using the time-invariance of the system. Now for the input-output trajectory $(\bar{u}_{[0,L-1]}, \bar{y}_{[0,L-1]})$ we can write the result from above in matrix notation like this:

$$\begin{bmatrix} \bar{y}(0) \\ \bar{y}(1) \\ \bar{y}(2) \\ \vdots \\ \bar{y}(L-1) \end{bmatrix} = \underbrace{\begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^{L-1} \end{bmatrix}}_{\Omega_L} x(0) + \underbrace{\begin{bmatrix} D & & & & \\ CB & D & & & \\ CAB & CB & D & & \\ \vdots & \ddots & \ddots & \ddots & \\ CA^{L-2}B & \dots & \dots & \dots & D \end{bmatrix}}_{\Theta_L} \begin{bmatrix} \bar{u}(0) \\ \bar{u}(1) \\ \bar{u}(2) \\ \vdots \\ \bar{u}(L-1) \end{bmatrix}$$

Matrix Ω_L is the classical observability matrix and Θ_L is the block Toeplitz matrix of, so called, Markov parameters. Now we can reformulate the last expression like this:

$$\begin{bmatrix} \bar{u}_{[0,L-1]} \\ \bar{y}_{[0,L-1]} \end{bmatrix} = \begin{bmatrix} 0 & I \\ \Omega_L & \Theta_L \end{bmatrix} \begin{bmatrix} x_{ini} \\ \bar{u}_{[0,L-1]} \end{bmatrix} \tag{2.2}$$

Since on the left side is a general input-output trajectory of length L (assuming time-invariance), we conclude that the space of all such trajectories is spanned by the columns of the $\begin{bmatrix} 0 & I \\ \Omega_L & \Theta_L \end{bmatrix}$ i.e. that space is the image of that matrix. It can be simply shown by rank-nullity theorem that:

$$\text{rank} \begin{bmatrix} 0 & I \\ \Omega_L & \Theta_L \end{bmatrix} = dim\ im \begin{bmatrix} 0 & I \\ \Omega_L & \Theta_L \end{bmatrix} = \text{rank}(\Omega_L) + mL$$

The columns of the Hankel matrix $\begin{bmatrix} H_L(u_{[0,T-1]}) \\ H_L(y_{[0,T-1]}) \end{bmatrix}$ are input-output trajectories so we conclude that:

$$im \begin{bmatrix} H_L(u_{[0,T-1]}) \\ H_L(y_{[0,T-1]}) \end{bmatrix} \subseteq im \begin{bmatrix} 0 & I \\ \Omega_L & \Theta_L \end{bmatrix}$$

To prove the Lemma we need to prove the equality of those spaces. We assume the strict inclusion:

$$im \begin{bmatrix} H_L(u_{[0,T-1]}) \\ H_L(y_{[0,T-1]}) \end{bmatrix} \subsetneq im \begin{bmatrix} 0 & I \\ \Omega_L & \Theta_L \end{bmatrix}$$

The expression (2.2) can be used to formulate an analog expression using Hankel matrices like this:

$$\begin{bmatrix} H_L(u_{[0,T-1]}) \\ H_L(y_{[0,T-1]}) \end{bmatrix} = \begin{bmatrix} 0 & I \\ \Omega_L & \Theta_L \end{bmatrix} \begin{bmatrix} X_{[0,T-L]} \\ H_L(u_{[0,T-1]}) \end{bmatrix}$$

By our assumption the matrix $J_L := \begin{bmatrix} X_{[0,T-L]} \\ H_L(u_{[0,T-1]}) \end{bmatrix}$ must not have full row rank. That implies that the left kernel of J_L is non-trivial

$\square$

References

[1] Author Name. *Title of the Book or Article*. Publisher or Journal, year.

A Additional Material